Protecting networks and data; protocols

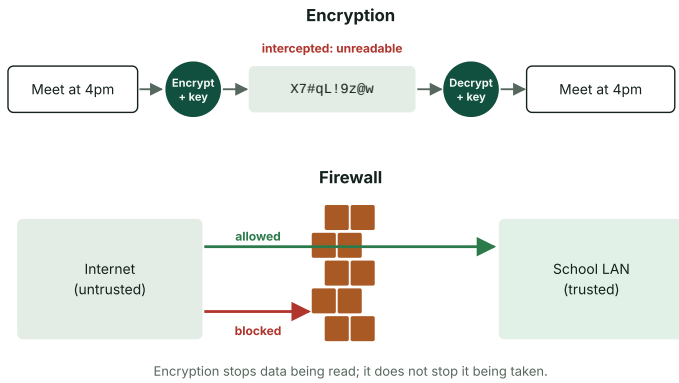
L19 FACT FILE

Specification: students should be able to

- Explain how networks and data can be protected using encryption, passwords, levels of access, backup and firewalls
- Describe the role of a protocol in data transfer
- Describe the purpose of the following protocols: File Transfer Protocol (FTP), HyperText Transfer Protocol (HTTP) and HyperText Transfer Protocol Secure (HTTPS)

CCEA fact file: Network Security and Data Transfer

Key words: encryption, decryption, encryption key, user ID, strong password, two-factor authentication, levels of access, read only, read and write, backup, restore, firewall, protocol, TCP/IP, FTP, HTTP, HTTPS, status code



Encryption makes intercepted data unreadable; a firewall filters network traffic

Protecting a network

Every organisation that stores data, from a primary school to a hospital trust, must stop that data being lost, stolen, read or changed by the wrong people. Under UK GDPR and the Data Protection Act 2018, organisations must keep personal data secure. The CCEA specification names five ways to protect networks and data: encryption, passwords, levels of access, backup and firewalls. Each one protects against a different problem. Passwords and levels of access control who can get in and what they can do once inside. Firewalls guard the boundary between the network and the internet. Encryption makes data useless to anyone who intercepts or steals it. Backups make sure data can be recovered if it is lost or corrupted. No single method is enough on its own, so organisations use them together. Data travelling across networks also follows protocols, the agreed rules that let different devices communicate, and some protocols, such as HTTPS, include encryption.

Encryption

Data sent across a network can be intercepted, and laptops and memory sticks can be lost or stolen. Encryption protects against this by using software, an algorithm and a key to scramble the data into cipher text before it is stored or sent. Anyone who intercepts the cipher text cannot understand it. Only someone with the correct key can decrypt it, turning it back into readable plain text. A simple example is a Caesar cipher, where each letter is shifted a set number of places along the alphabet, but real encryption uses very long keys and complex mathematics so it cannot be cracked by guessing. Encryption is used to secure Wi-Fi networks, HTTPS websites, messaging apps and staff laptops. A common exam mistake is to say encryption stops data being intercepted or stolen. It does not; it means that if the data is intercepted or stolen, it cannot be read or understood.

Passwords and levels of access

Each network user is given a unique user ID and a password. When they log on, both are checked against a secure database of users; if they do not match, access is refused. Passwords are shown as asterisks so others cannot read them, accounts can be locked after several failed attempts, and users may be made to change passwords regularly. A strong password is long, mixes upper and lower case letters, digits and symbols, and is not a dictionary word or personal detail such as a pet's name. Two-factor authentication adds a second check, for example a code sent to a phone. Levels of access then control what each user can do once logged on: no access, read only, read and copy, or read and write. Rights are stored against each user ID and checked whenever a file is opened. This keeps confidential data private, prevents accidental or deliberate changes, and limits the damage if one account is hacked.

Backups and firewalls

A backup is a copy of data stored separately from the original so that it can be restored if the original is lost, corrupted, deleted or encrypted by ransomware. Network backups are usually scheduled to run automatically, often overnight, and are kept in a safe place away from the server, for example on an external drive kept off-site or in cloud storage, so that a fire or theft does not destroy both copies. When needed, restore software copies the data back. A firewall is hardware or software that sits between a network and the internet. It monitors and filters all data entering or leaving the network and checks it against a set of rules called the security policy. Data that meets the rules is allowed through; data that does not is blocked. This keeps hackers out, can stop some viruses and spam, and can stop users visiting unsuitable websites. Firewalls also log blocked attempts so the network manager can investigate.

Protocols: FTP, HTTP and HTTPS

A protocol is an agreed set of rules for sending and receiving data. Sender and receiver must use the same protocol, rather like two people needing a shared language. Protocols set out the format of the data, how it is addressed, how errors are checked and how communication begins and ends, so devices from different manufacturers can work together. The internet uses TCP/IP. FTP, the File Transfer Protocol, is used to upload and download files between computers and a file server, for example when a web designer uploads a club's website files; users normally log in with a username and password. HTTP, the HyperText Transfer Protocol, is used by browsers to request web pages from web servers; if a page is missing the server returns a status code such as 404. HTTPS is HTTP with encryption, so intercepted data such as card numbers is meaningless. Browsers show a padlock for HTTPS sites.

T Key terms from the mark schemes

Encryption
Scrambling data using an algorithm and a key so that it cannot be understood by anyone who does not have the key.
`Credited: Encoding data so it is unreadable if intercepted`

Decryption
Turning encrypted data back into readable form using the correct key.

Firewall
Hardware or software that monitors and filters data entering or leaving a network, blocking traffic that does not meet security rules.

Access rights / levels of access
Permissions linked to a user ID that control what each user can see and do, e.g. no access, read only, read and write.
`Credited: Access levels`

Read only
An access level where a user can view data but cannot change it.

Strong password
A long password that mixes upper and lower case letters, digits and symbols and is not a dictionary word or personal detail.

Two-factor authentication
A logon that needs two different forms of verification, such as a password and a code sent to a phone.
`Credited: 2FA`

Backup
A copy of data stored separately from the original so it can be restored if the original is lost or corrupted.

Protocol
An agreed set of rules governing how data is transmitted between devices.
`Credited: Set of rules for communication`

FTP
File Transfer Protocol: rules for uploading and downloading files between computers over a network.

HTTP
HyperText Transfer Protocol: rules a browser uses to request web pages from a web server and transfer them.

HTTPS
HyperText Transfer Protocol Secure: HTTP with encryption, so data sent between browser and server cannot be read if intercepted.

Exam focus

This lesson is tested almost every year. Encryption appeared in 2021 Q10, 2022 Q12, 2023 Q8 and 2024 Q6; firewalls in 2022 Q12; strong passwords in 2021 Q9; and protocols, FTP and HTTPS in 2021 Q9, 2022 Q7 and 2023 Q8. 2024 Q11 asked for protection methods in context, and terms such as protocol and firewall turn up in matching questions. For encryption, two marks usually need what it does (scrambles data so it cannot be understood) and who can read it (only someone with the key). Never say encryption stops data being stolen. For a protocol, say it is a set of rules and give a detail, such as allowing different devices to communicate. When asked to 'expand' an acronym, spell every word correctly. For HTTP versus HTTPS, the key point is that HTTPS encrypts the data, so you must compare both sides. Always link protection methods to the scenario given.

`Unit 1 2024 Q11`	`[6]`	Legislation matching, online harms, protection methods
`Unit 1 2024 Q10`	`[8]`	Key term matching
`Unit 1 2024 Q6`	`[13]`	Network statements, bus disadvantages, LAN, encryption, HTTPS, RAM vs ROM, malware routes
`Unit 1 2023 Q8`	`[9]`	Protocols, HTTPS, phishing, malware, encryption
`Unit 1 2022 Q12`	`[7]`	WAN, encryption, firewall, processor performance

? Check yourself

1. Explain how encryption protects data. [2]
2. Describe the purpose of a firewall. [2]
3. Explain how levels of access protect patient data in a hospital. [2]

Answers: 1. Data is scrambled/encoded using a key/algorithm (1) / so if it is intercepted or stolen it cannot be understood; only someone with the key can decrypt it (1). 2. Monitors/filters data entering and leaving a network (1) / blocks traffic that does not meet the security rules, e.g. stops hackers gaining access from the internet (1). 3. Staff are given access only to the data they need for their job, e.g. receptionists see appointments but not medical notes (1) / so confidential patient data is not seen or changed by unauthorised staff; read-only access prevents records being altered (1).